Mini/Micro LED Design and Challenges for Advanced Displays

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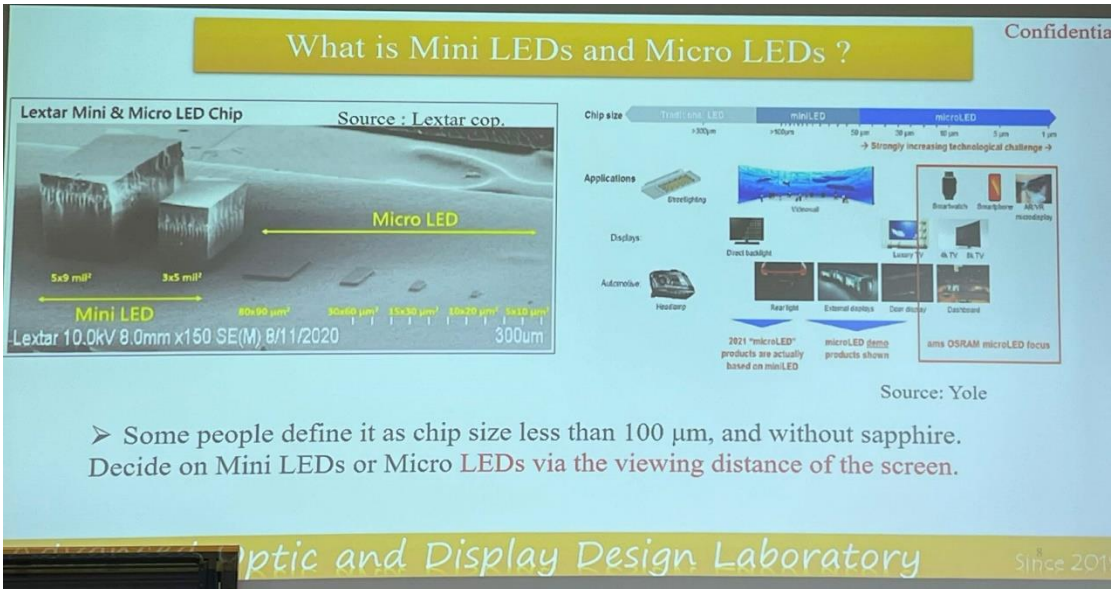
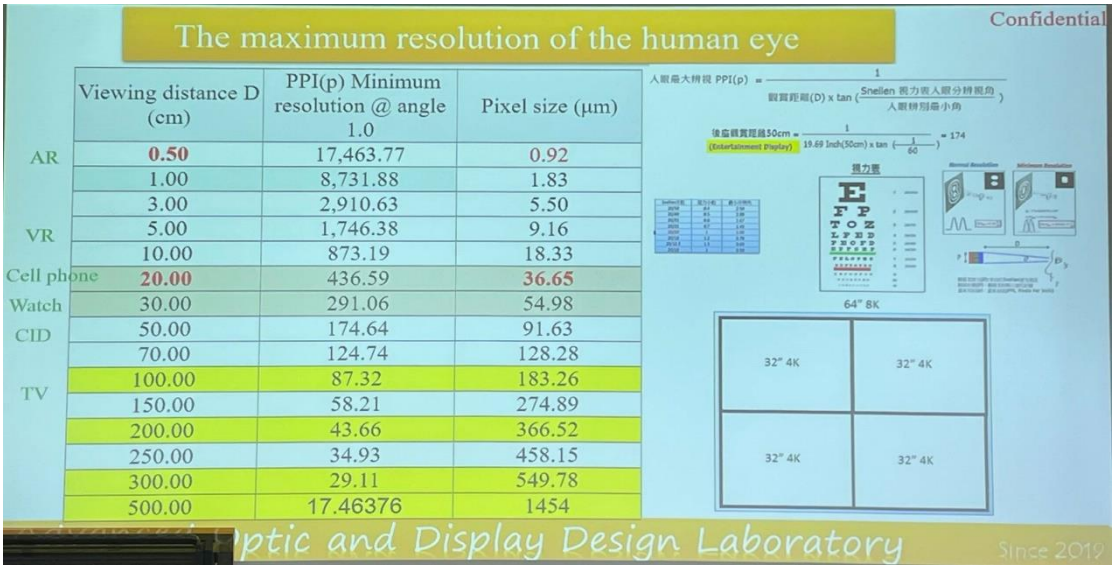
筆記：

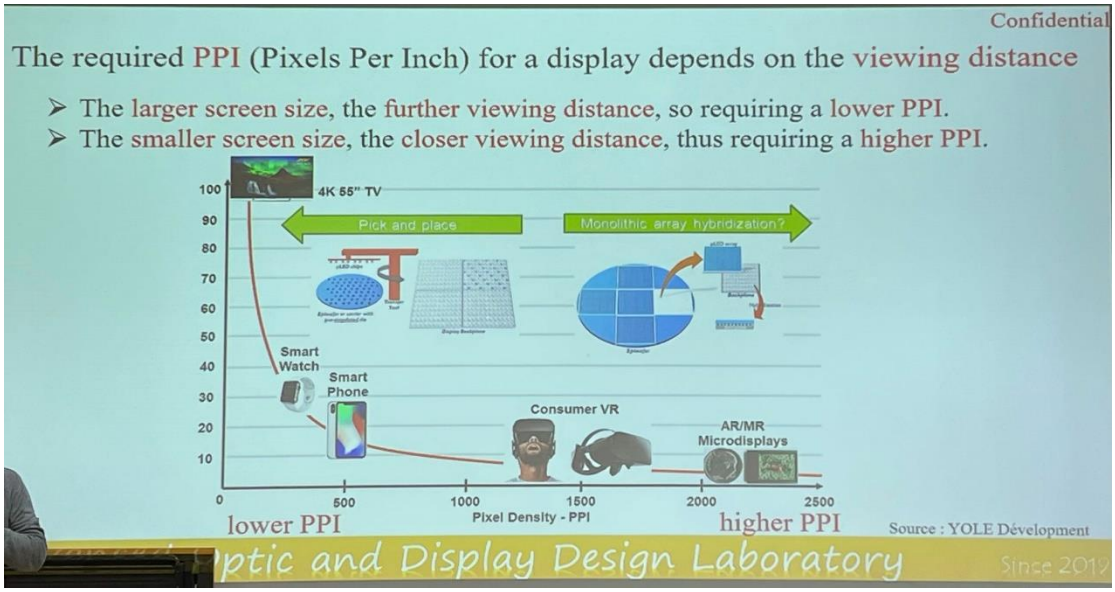
What is Mini LEDs and Micro LEDs ?



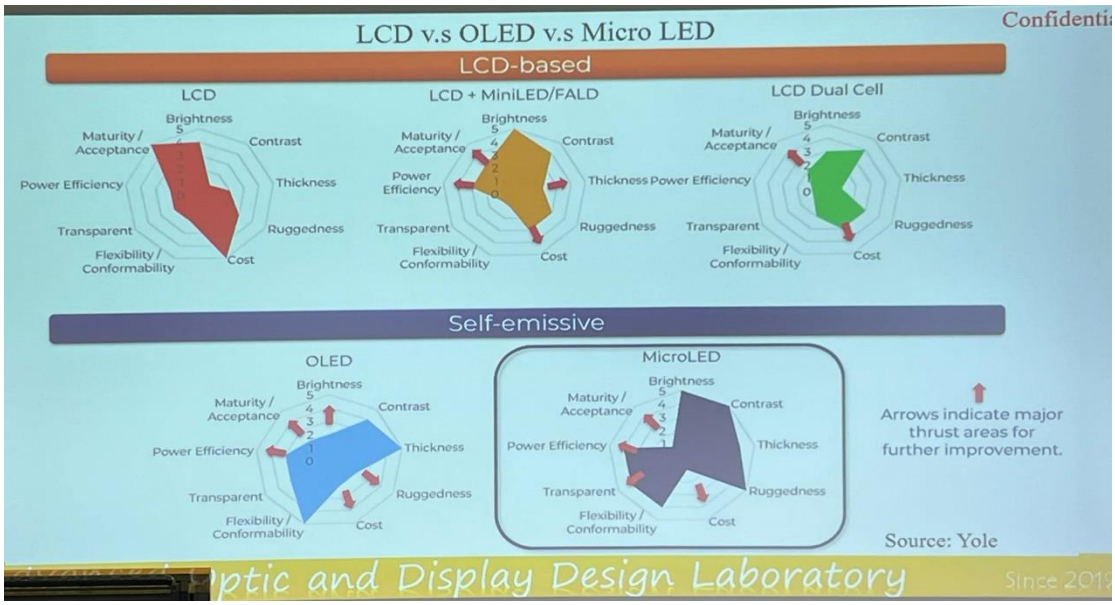
The maximum resolution of the human eyes



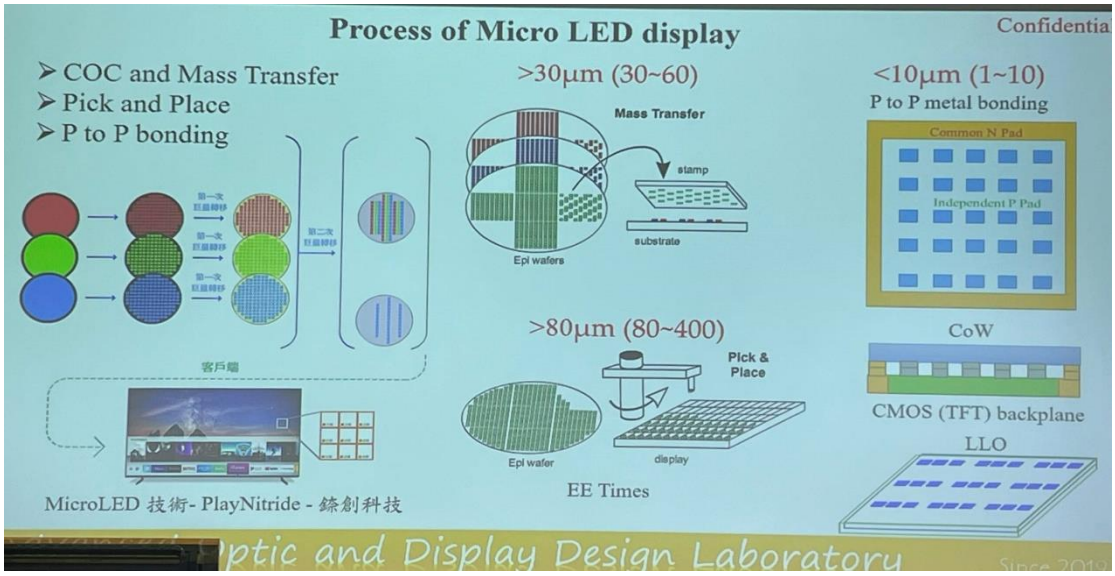
	Viewing distance D (cm)	PPI(p) Minimum resolution @ angle 1.0	Pixel size (μm)
AR	0.50	17,463.77	0.92
	1.00	8,731.88	1.83
	3.00	2,910.63	5.50
VR	5.00	1,746.38	9.16
	10.00	873.19	18.33
Cell phone	20.00	436.59	36.65
Watch	30.00	291.06	54.98
CID	50.00	174.64	91.63
	70.00	124.74	128.28
TV	100.00	87.32	183.26
	150.00	58.21	274.89
	200.00	43.66	366.52
	250.00	34.93	458.15
	300.00	29.11	549.78
	500.00	17.46376	1454

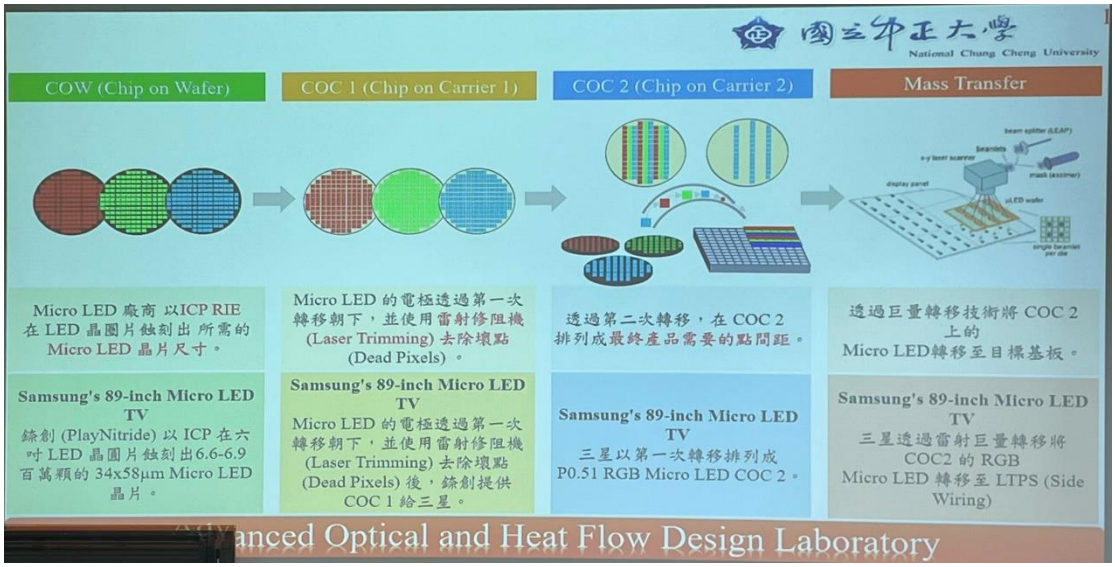


LCD vs OLED vs Micro LED

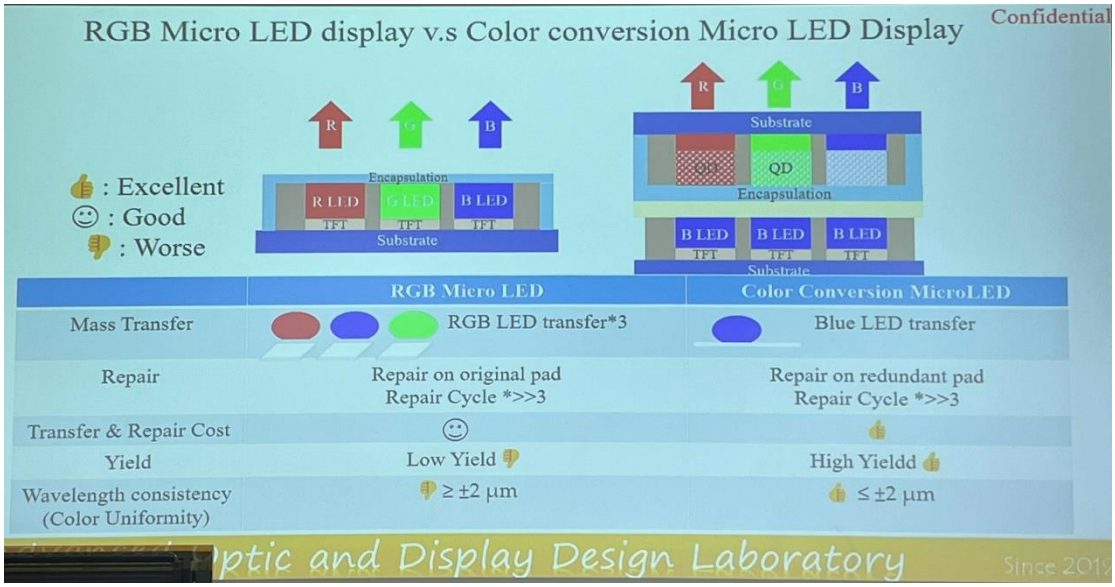


Process of Micro LED display

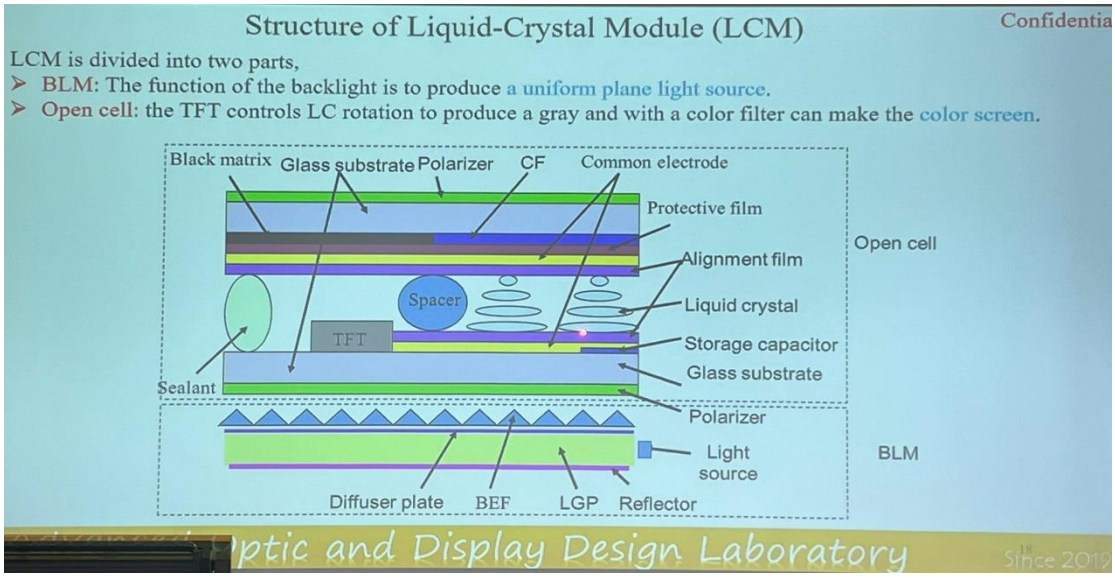




RGB Micro LED display vs Color conversion Micro LED display

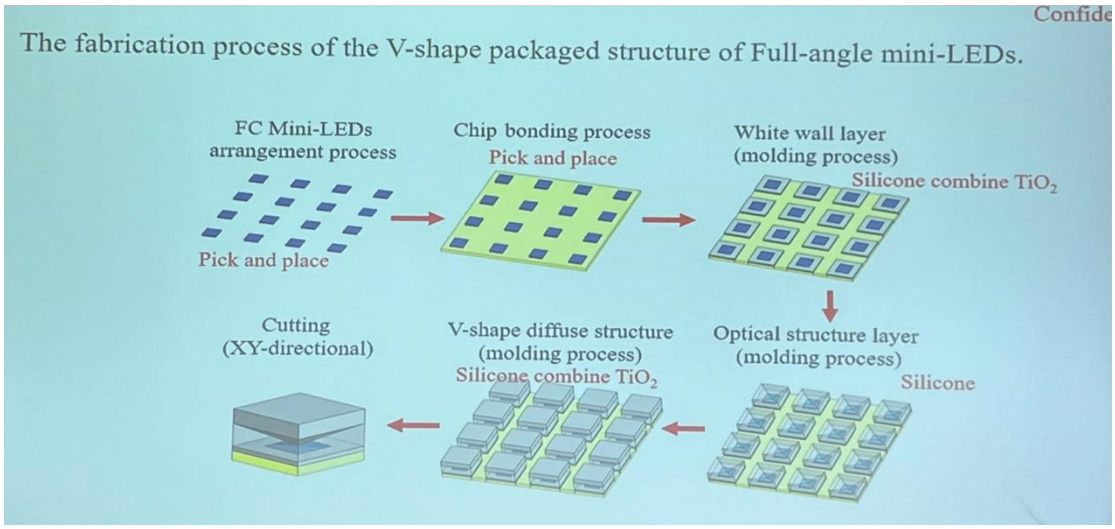


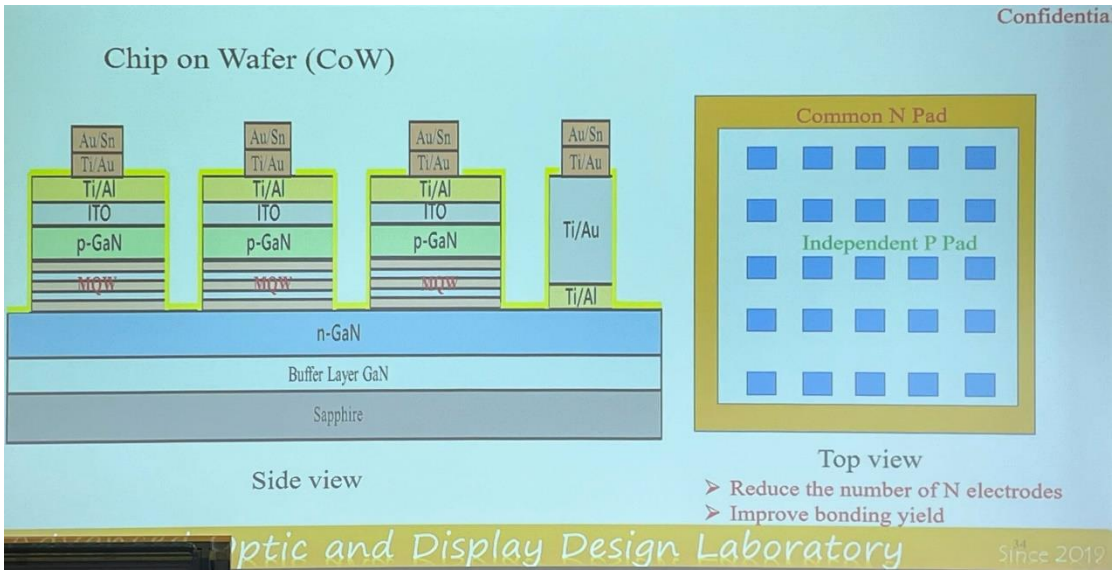
Structure of Liquid-Crystal Module (LCM)



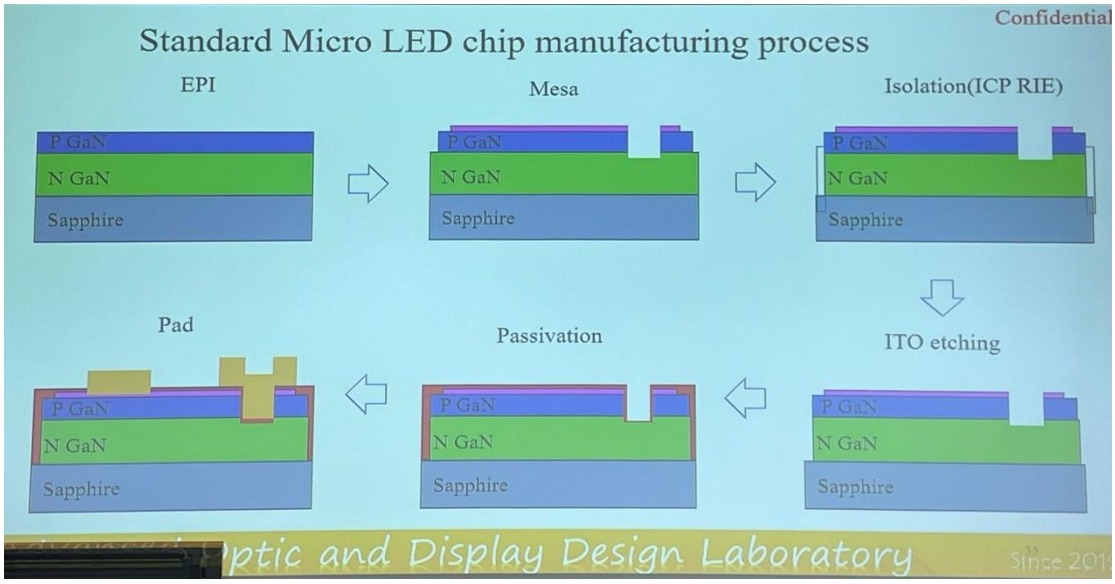
最下層光來源透光率從 100%到最上層下降至 40-60%。

The fabrication process of the V-shape packaged structure of Full-angle mini-LEDs

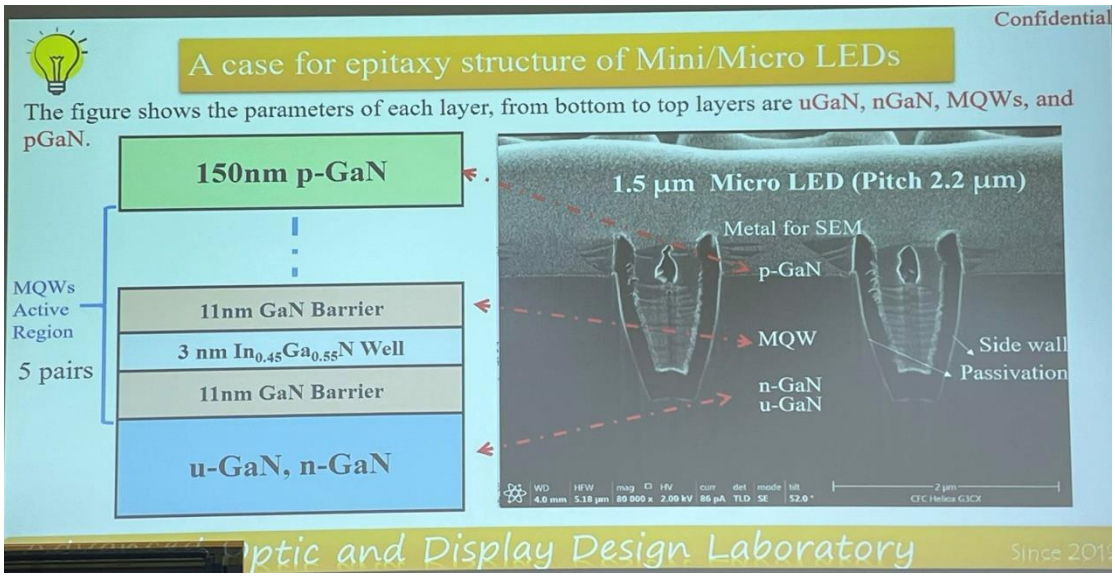


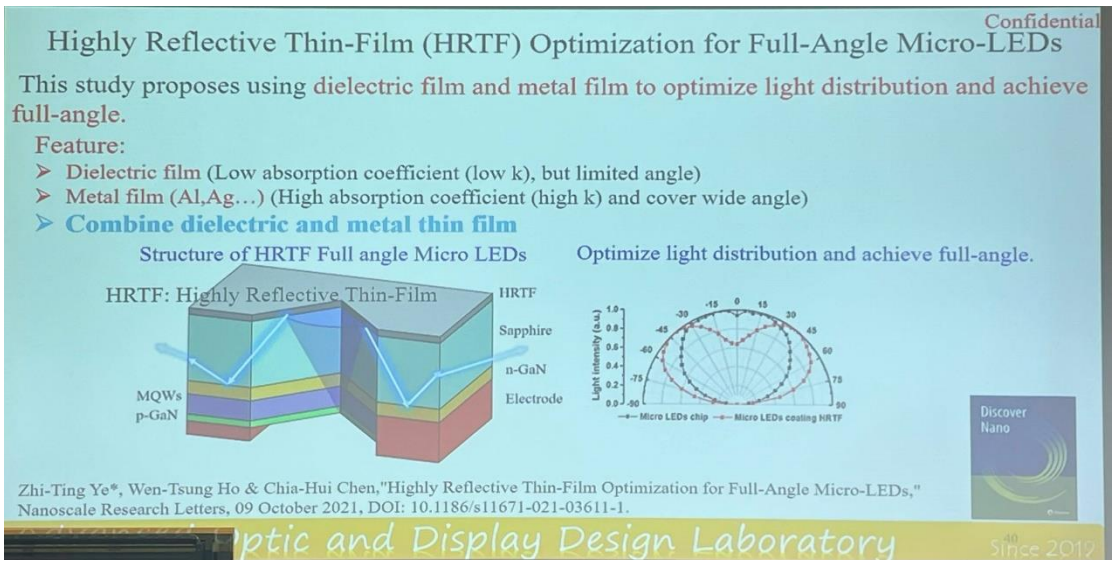


Standard Micro LED chip manufacturing process



A case for epitaxy structure of Mini/Micro LEDs



Highly Reflective Thin-Film (HRTF) Optimization

for Full-angle Micro-LEDs